

$$22) \{3\}11\bar{1}2 \rightarrow$$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{9} g_2^4 LF_{3,0}[m_2] \delta_{i1i2} + \frac{1}{6} g_2^4 LF_{4,-1}[m_2] \delta_{i1i2} - \frac{8}{45} g_2^4 LF_{5,-2}[m_2] \delta_{i1i2} + \frac{1}{18} \sum_p g_2^4 LF_{3,0}[m_l^p] \right. \\ & \delta_{i1i2} - \frac{5}{48} \sum_p g_2^4 LF_{4,-1}[m_l^p] \delta_{i1i2} + \frac{2}{45} \sum_p g_2^4 LF_{5,-2}[m_l^p] \delta_{i1i2} + \frac{1}{6} \sum_p g_2^4 LF_{3,0}[m_q^p] \delta_{i1i2} - \\ & \frac{5}{16} \sum_p g_2^4 LF_{4,-1}[m_q^p] \delta_{i1i2} + \frac{2}{15} \sum_p g_2^4 LF_{5,-2}[m_q^p] \delta_{i1i2} + \frac{1}{18} g_2^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} + \\ & \frac{1}{12} g_2^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} - \frac{4}{45} g_2^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} + \frac{1}{2} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,1,0}[m_1, m_e^p] - \\ & g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{3,1,-1}[m_1, m_e^p] + \frac{1}{2} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{4,1,-2}[m_1, m_e^p] + \\ & \frac{1}{96} g_1^2 g_2^2 LF_{2,1,0}[m_1, m_l^{i1}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 LF_{2,2,-1}[m_1, m_l^{i1}] \delta_{i1i2} - \\ & \frac{1}{48} g_1^2 g_2^2 LF_{3,1,-1}[m_1, m_l^{i1}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 LF_{4,1,-2}[m_1, m_l^{i1}] \delta_{i1i2} + \\ & \frac{1}{96} g_1^2 g_2^2 LF_{2,1,0}[m_1, m_l^{i2}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 LF_{2,2,-1}[m_1, m_l^{i2}] \delta_{i1i2} - \\ & \frac{1}{48} g_1^2 g_2^2 LF_{3,1,-1}[m_1, m_l^{i2}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 LF_{4,1,-2}[m_1, m_l^{i2}] \delta_{i1i2} - \\ & \frac{1}{96} g_2^4 LF_{2,1,0}[m_2, m_l^{i1}] \delta_{i1i2} - \frac{1}{96} g_2^4 LF_{2,2,-1}[m_2, m_l^{i1}] \delta_{i1i2} - \frac{5}{48} g_2^4 LF_{3,1,-1}[m_2, m_l^{i1}] \delta_{i1i2} + \\ & \frac{1}{32} g_2^4 LF_{4,1,-2}[m_2, m_l^{i1}] \delta_{i1i2} - \frac{1}{96} g_2^4 LF_{2,1,0}[m_2, m_l^{i2}] \delta_{i1i2} - \frac{1}{96} g_2^4 LF_{2,2,-1}[m_2, m_l^{i2}] \delta_{i1i2} - \\ & \frac{5}{48} g_2^4 LF_{3,1,-1}[m_2, m_l^{i2}] \delta_{i1i2} + \frac{1}{32} g_2^4 LF_{4,1,-2}[m_2, m_l^{i2}] \delta_{i1i2} - \\ & \frac{1}{3} g_2^4 LF_{2,1,0}[m_2, \tilde{\mu}] \delta_{i1i2} - \frac{1}{3} g_2^4 LF_{2,2,-1}[m_2, \tilde{\mu}] \delta_{i1i2} + \frac{1}{6} g_2^4 LF_{3,1,-1}[m_2, \tilde{\mu}] \delta_{i1i2} + \\ & \frac{1}{6} g_2^4 LF_{3,2,-2}[m_2, \tilde{\mu}] \delta_{i1i2} + \frac{1}{8} g_2^2 \overline{a_e}^{pr} a_e^{pr} LF_{4,1,-1}[m_l^p, m_e^r] \delta_{i1i2} - \\ & \frac{1}{6} g_2^2 \overline{a_e}^{pr} a_e^{pr} LF_{5,1,-2}[m_l^p, m_e^r] \delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2 LF_{2,1,0}[m_l^{i1}, m_1] \delta_{i1i2} + \\ & \frac{1}{96} g_1^2 g_2^2 LF_{3,1,-1}[m_l^{i1}, m_1] \delta_{i1i2} + \frac{1}{48} g_2^4 LF_{2,1,0}[m_l^{i1}, m_2] \delta_{i1i2} - \\ & \frac{1}{96} g_2^4 LF_{3,1,-1}[m_l^{i1}, m_2] \delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2 LF_{2,1,0}[m_l^{i2}, m_1] \delta_{i1i2} + \\ & \frac{1}{96} g_1^2 g_2^2 LF_{3,1,-1}[m_l^{i2}, m_1] \delta_{i1i2} + \frac{1}{48} g_2^4 LF_{2,1,0}[m_l^{i2}, m_2] \delta_{i1i2} - \\ & \frac{1}{96} g_2^4 LF_{3,1,-1}[m_l^{i2}, m_2] \delta_{i1i2} + \frac{3}{8} g_2^2 \overline{a_d}^{pr} a_d^{pr} LF_{4,1,-1}[m_q^p, m_d^r] \delta_{i1i2} - \\ & \frac{1}{2} g_2^2 \overline{a_d}^{pr} a_d^{pr} LF_{5,1,-2}[m_q^p, m_d^r] \delta_{i1i2} + \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{2,2,0}[m_q^p, m_u^r] \delta_{i1i2} - \\ & \frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{3,2,-1}[m_q^p, m_u^r] \delta_{i1i2} - \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{3,1,0}[m_u^r, m_q^p] \delta_{i1i2} - \\ & \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{3,2,-1}[m_u^r, m_q^p] \delta_{i1i2} + \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{4,1,-1}[m_u^r, m_q^p] \delta_{i1i2} - \\ & \frac{1}{2} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{5,1,-2}[m_u^r, m_q^p] \delta_{i1i2} - \frac{5}{24} g_1^2 g_2^2 LF_{4,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 g_2^2 LF_{5,1,-3}[\tilde{\mu}, m_1] \delta_{i1i2} + \frac{2}{3} g_2^4 LF_{3,1,-1}[\tilde{\mu}, m_2] \delta_{i1i2} + \frac{1}{3} g_2^4 LF_{3,2,-2}[\tilde{\mu}, m_2] \delta_{i1i2} - \\ & \frac{9}{8} g_2^4 LF_{4,1,-2}[\tilde{\mu}, m_2] \delta_{i1i2} + \frac{1}{2} g_2^4 LF_{5,1,-3}[\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{8} g_2^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{3,1,-1}[\tilde{\mu}, m_e^p] + \\ & \frac{1}{24} g_2^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{4,1,-2}[\tilde{\mu}, m_e^p] + \frac{1}{2} \overline{y_e}^{rp} \overline{y_e}^{i2s} y_e^{rs} y_e^{i1p} LF_{2,1,0}[\tilde{\mu}, m_l^r] - \\ & \overline{y_e}^{rp} \overline{y_e}^{i2s} y_e^{rs} y_e^{i1p} LF_{3,1,-1}[\tilde{\mu}, m_l^r] + \frac{1}{2} \overline{y_e}^{rp} \overline{y_e}^{i2s} y_e^{rs} y_e^{i1p} LF_{4,1,-2}[\tilde{\mu}, m_l^r] - \\ & \frac{1}{8} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,1,1,0}[m_1, m_e^p, m_l^{i1}] - \frac{1}{16} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,2,1,-1}[m_1, m_e^p, m_l^{i1}] + \\ & \frac{1}{8} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{3,1,1,-1}[m_1, m_e^p, m_l^{i1}] - \frac{1}{8} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{2,1,1,0}[m_1, m_e^p, m_l^{i2}] - \\ & \frac{1}{16} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{2,2,1,-1}[m_1, m_e^p, m_l^{i2}] + \frac{1}{8} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{3,1,1,-1}[m_1, m_e^p, m_l^{i2}] - \\ & \frac{3}{4} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,1,1,-1}[m_1, m_e^p, \tilde{\mu}] + \frac{1}{2} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{3,1,1,-2}[m_1, m_e^p, \tilde{\mu}] + \\ & \frac{1}{16} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,2,1,-1}[m_1, m_l^{i1}, m_e^p] + \frac{1}{16} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{2,2,1,-1}[m_1, m_l^{i2}, m_e^p] + \\ & \frac{5}{16} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_e^p] - \frac{1}{16} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_l^{i1}] - \\ & \frac{1}{16} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_l^{i2}] + \frac{1}{16} g_2^4 LF_{2,1,1,-1}[m_2, m_l^{i1}, \tilde{\mu}] \delta_{i1i2} - \\ & \frac{1}{8} g_2^4 m_2^2 LF_{2,1,1,0}[m_2, m_l^{i1}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{16} g_2^4 LF_{2,1,1,-1}[m_2, m_l^{i2}, \tilde{\mu}] \delta_{i1i2} - \\ & \frac{1}{8} g_2^4 m_2^2 LF_{2,1,1,0}[m_2, m_l^{i2}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{16} g_2^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_e^p] + \\ & \frac{1}{16} (3 g_2^2 \overline{y_e}^{i2p} y_e^{i1p} + g_2^4 \delta_{i1i2}) LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_l^{i1}] + \\ & \frac{1}{16} g_2^4 m_2^2 LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^{i1}] \delta_{i1i2} + \frac{1}{16} (3 g_2^2 \overline{y_e}^{i2p} y_e^{i1p} + g_2^4 \delta_{i1i2}) \\ & LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_l^{i2}] + \frac{1}{16} g_2^4 m_2^2 LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^{i2}] \delta_{i1i2} + \\ & \frac{1}{16} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_e^p] + \frac{3}{16} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_l^{i1}] - \\ & \frac{1}{8} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_l^{i1}] + \frac{3}{16} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_l^{i2}] - \\ & \frac{1}{8} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_l^{i2}] - \frac{1}{16} g_2^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_e^p] - \\ & \frac{9}{16} g_2^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_l^{i1}] + \frac{3}{8} g_2^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{3,1,1,-2}[\tilde{\mu}, m_2, m_l^{i1}] - \\ & \frac{9}{16} g_2^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_l^{i2}] + \frac{3}{8} g_2^2 \overline{y_e}^{i2p} y_e^{i1p} LF_{3,1,1,-2}[\tilde{\mu}, m_2, m_l^{i2}] - \\ & \frac{1}{16} g_1^2 g_2^2 LF_{1,1,1,1,-1}[m_1, m_2, m_l^{i1}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{8} m_1 m_2 g_1^2 g_2^2 LF_{1,1,1,1,0}[m_1, m_2, m_l^{i1}, \tilde{\mu}] \delta_{i1i2} - \\ & \frac{1}{16} g_1^2 g_2^2 LF_{1,1,1,1,-1}[m_1, m_2, m_l^{i2}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{8} m_1 m_2 g_1^2 g_2^2 LF_{1,1,1,1,0}[m_1, m_2, m_l^{i2}, \tilde{\mu}] \delta_{i1i2} + \\ & \frac{1}{16} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{1,1,1,1,0}[m_1, m_e^p, m_l^{i1}, \tilde{\mu}] + \frac{1}{32} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} \\ & LF_{2,1,1,1,-1}[m_1, m_e^p, m_l^{i1}, \tilde{\mu}] + \frac{1}{16} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{1,1,1,1,0}[m_1, m_e^p, m_l^{i2}, \tilde{\mu}] + \\ & \frac{1}{32} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{2,1,1,1,-1}[m_1, m_e^p, m_l^{i2}, \tilde{\mu}] + \frac{1}{16} m_2 g_2^2 \overline{y_e}^{i2p} a_e^{i1p} \\ & LF_{1,1,1,1,0}[m_2, m_e^p, m_l^{i1}, \tilde{\mu}] + \frac{1}{32} m_2 g_2^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,1,1,1,-1}[m_2, m_e^p, m_l^{i1}, \tilde{\mu}] + \\ & \frac{1}{16} m_2 g_2^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{1,1,1,1,0}[m_2, m_e^p, m_l^{i2}, \tilde{\mu}] + \frac{1}{32} m_2 g_2^2 y_e^{i1p} \overline{a_e}^{i2p} \\ & LF_{2,1,1,1,-1}[m_2, m_e^p, m_l^{i2}, \tilde{\mu}] - \frac{1}{16} g_1^2 \overline{a_e}^{i2p} a_e^{i1p} LF_{2,1,1,1,-1}[m_e^p, m_1, m_l^{i1}, m_l^{i2}] - \\ & \frac{1}{32} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,1,1,1,-1}[m_e^p, m_1, m_l^{i1}, \tilde{\mu}] - \frac{1}{32} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} \\ & LF_{2,1,1,1,-1}[m_e^p, m_1, m_l^{i2}, \tilde{\mu}] + \frac{1}{16} g_2^2 \overline{a_e}^{i2p} a_e^{i1p} LF_{2,1,1,1,-1}[m_e^p, m_2, m_l^{i1}, m_l^{i2}] - \\ & \frac{1}{32} m_2 g_2^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,1,1,1,-1}[m_e^p, m_2, m_l^{i1}, \tilde{\mu}] - \frac{1}{32} m_2 g_2^2 y_e^{i1p} \overline{a_e}^{i2p} \\ & LF_{2,1,1,1,-1}[m_e^p, m_2, m_l^{i2}, \tilde{\mu}] + \frac{1}{32} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,1,1,1,-1}[m_l^{i1}, m_1, m_e^p, \tilde{\mu}] + \\ & \frac{1}{32} m_2 g_2^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,1,1,1,-1}[m_l^{i1}, m_2, m_e^p, \tilde{\mu}] + \frac{1}{32} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} \\ & LF_{2,1,1,1,-1}[m_l^{i2}, m_1, m_e^p, \tilde{\mu}] + \frac{1}{32} m_2 g_2^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{2,1,1,1,-1}[m_l^{i2}, m_2, m_e^p, \tilde{\mu}] - \\ & \frac{1}{16} g_1^2 g_2^2 LF_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_l^{i1}] \delta_{i1i2} + \frac{1}{16} m_1 m_2 g_1^2 g_2^2 LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_l^{i1}] \\ & \delta_{i1i2} - \frac{1}{16} g_1^2 g_2^2 LF_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_l^{i2}] \delta_{i1i2} + \frac{1}{16} m_1 m_2 g_1^2 g_2^2 \\ & LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_l^{i2}] \delta_{i1i2} - \frac{1}{32} m_1 g_1^2 \overline{y_e}^{i2p} a_e^{i1p} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_e^p, m_l^{i1}] - \\ & \frac{1}{32} m_1 g_1^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_e^p, m_l^{i2}] - \frac{1}{32} m_2 g_2^2 \overline{y_e}^{i2p} a_e^{i1p} \\ & LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_e^p, m_l^{i1}] - \frac{1}{32} m_2 g_2^2 y_e^{i1p} \overline{a_e}^{i2p} LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_e^p, m_l^{i2}] \Big) \end{aligned}$$